Reflection and damping behavior of signal propagation in coaxial cables

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We observed different properties of electromagnetic waves in a coaxial cable. One can see the influence of different resistors at the ends of the cable on the reflection. This was used to determine the characteristic impendance of the used cable. We measured the propagation velocity of waves in a coaxial cable via a runtime measurent and via standing waves. With that one observes a frequency dependent damping coefficient of the cable. In addition we determined the speed of light in air $c_{\text{air}} = (2.9421 \pm 0.1205) \cdot 10^8 \, \frac{\text{m}}{\text{s}}$ using a runtime measurement.

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1 Introduction

It is a general goal in many fields to transmit signals with a minimum loss. One simple way to achieve this is using a coaxial cable. Due to this property coaxial cables appear in many applications such as laboratorium technique or TV technique.
Therefore it is important to examine how these cables transmit signals. We will cover the impendance, the propagation velocity and the damping inside a coaxial cable. Furthermore there will be a measurement to determine the speed of light in air.

2 Coaxial cable

A coaxial cable is a transmission line used to transfer high-frequency electrical signals. The signals are not transmitted via electric current but electromagnetic radiation. In order to enable this transmission with low losses, the signal conductor is surrounded by a concentric conducting shield. The inner conductor, which carries the signal, usually consists of a copper wire. The space between the inner and outer conductor is filled with a dielectric and the cylindrical outer conductor is again surrounded by an insulator (Fig. 1).
The following wave equations apply to current and voltage curves in a homogeneous, loss-free conductor

$$\frac{\partial^2 U}{\partial x^2} = L^* C^* \frac{\partial^2 U}{\partial t^2} \tag{1}$$

$$\frac{\partial^2 I}{\partial x^2} = L^* C^* \frac{\partial^2 I}{\partial t^2} \tag{2}$$

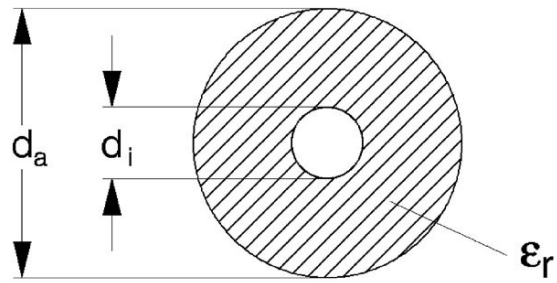

Fig. 1: Schematic cross-section of a coaxial cable with inner conductor diameter d_i and outer conductor diameter d_a. Between the two conductors there is an insulator with the relative dielectric constant ϵ_r

where L^* is the inductance per unit length and C^* is the capacity per unit length. From these wave equations one obtains the value for the propagation velocity:

$$v = \frac{1}{\sqrt{L^* C^*}} \tag{3}$$

The propagation behavior of a signal in an ideal coaxial cable depends only on L^* and C^*. If you rewrite L^* and C^* you get

$$v = \frac{c}{\sqrt{\mu_\text{r} \epsilon_\text{r}}} \tag{4}$$

where μ_r is the magnetic permeability and $c = 299792458 \, \frac{\text{m}}{\text{s}}$ is the speed of light in vacuum. The ratio between voltage and current at all points of the

coaxial cable is described by the characteristic impedance Z.

$$Z = \sqrt{\frac{L^*}{C^*}} \tag{5}$$

Although Z has the same unit as an ohmic resistance, those two aren't related since Z doesn't consume any power.
The reflection factor ρ shows the ratio of the wave reflected at the end of a conductor to the incoming wave.

$$\rho = \frac{R_\mathrm{V} - Z}{R_\mathrm{V} + Z} \tag{6}$$

R_V is the ohmic resistance of the appliance at the end of the cable.
Three specific cases can be considered:
If the coaxial cable has an open end, the signal is reflected without a phase shift. Thus the voltage detectable at the end is twice as large as the applied one.
If the cable is short-circuited, the reflection coefficient is $\rho = -1$ since $R_\mathrm{V} = 0$ and the total voltage at the end of the cable is zero.
In order to prevent reflections occuring at the end of the cable, $Z = R_\mathrm{V}$ must apply to the appliance resistance, since ρ then assumes the value zero.

3 Experimental setup

To perform the measurements with the coaxial cable, the experiments were set up according to Fig. 2.

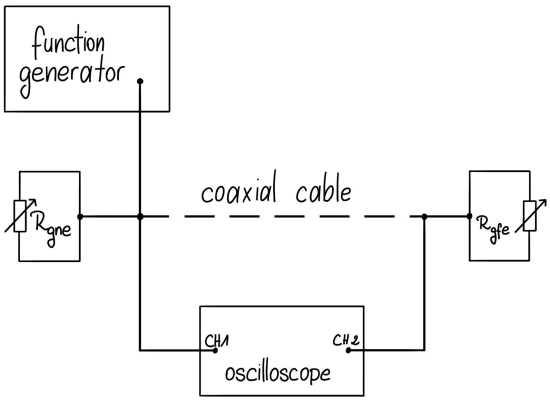

Fig. 2: Circuit diagram of the experimental setup. An oscilloscope plots the signal generated by a function generator at both ends of a coaxial cable.

The function generator *Agilent* 33220A is attached to a junction box to which the generator near end of the cable (gne) and a variable resistor R_gne are connected. The function generator has an output impedance of $R_\mathrm{out} = 50\,\Omega$ which is the reason why no reflections occur at the output. The generator far end of the cable (gfe) is combined with another junction box, to which a variable resistor R_gfe is attached. Both of the junction boxes are combined with the oscilloscope *Tektronix* TDS 1001B. The junction box combined with the generator near end of the cable is connected to channel 1 and the junction box combined with the generator far end is connected to channel 2. The oscilloscope has a high input impedance, resulting in the cable being nearly open ended when connected to the oscilloscope.

4 Impedance of a coaxial cable

In order to determine the impendance of the coaxial cable, the resistance R_gfe was increased until no reflection of the pulse was detected at the beginning of the cable. No resistance R_gne was used in this measurement. Therefore the cable has an open end at the generator near end. $R_\mathrm{gfe} = Z$ applies for the resistance R_gfe when the reflected pulse vanishes. For this measurement three different variable resistors were used. The results are shown in Tab. 1. The measured

resistor	resistance R_gfe [Ω]
carbon film potentiometer	49.9 ± 0.9
wire wound potentiometer	48.0 ± 1.0
resistance decade	50.0 ± 1.0

Tab. 1: Resistance R_gfe of different variable resistors for a vanishing reflected puls.

resistances coincide within the limits of their uncertainty. The impedance Z of the cable is determined to $Z = (49.34 \pm 0.56)\,\Omega$ using the mean value of the three resistances. This value coincides withe the given impedance $Z = 50\,\Omega$ for coaxial cabels.
A commercial $50\,\Omega$ plug used as R_gfe confirms this result as the reflected pulse vanishes. In addition, the end of the cable was short-circuited and the total voltage at the beginning of the cable reduced to zero.

5 Wave forms in a coaxial cable

5.1 Reflected pulses

The function generator emitts voltage pulses with an amplitude of 10 V and a width of 20 ns with a frequency of 200 KHz. In order to observe the effect of different termination impedance at the beginning and the end of the cable, nine different combinations of R_gne and R_gfe are connected.
First the commercial $50\,\Omega$ resistor was connected to the generator near end and the three different resistors $R \in \{0, 50\,\Omega, \infty\}$ were used for the generator

far end. With a resistance $R_{\text{gfe}} \approx 50\,\Omega$, the amplitude of the pulse becomes smaller due to the damping of the cable and is hardly reflected at the end of the cable. Therefore the pulse at the end of the cable is smaller than at the beginning and only a small reflected signal is detected again at the beginning of the cable (Fig. 3). The small reflected pulse occurs, because R_{gfe} does not correspond exactly to the impedance Z of the cable.

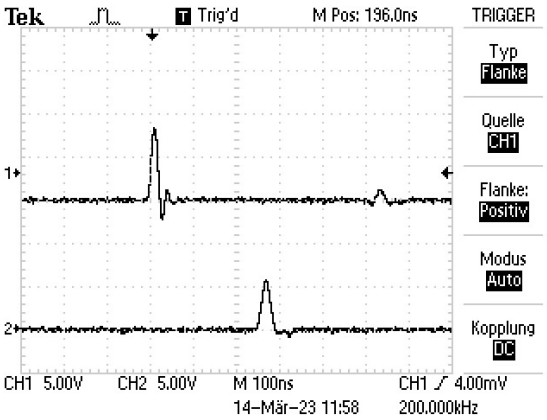

Fig. 3: Detected pulses at both ends of the cable with $R_{\text{gne}} = 50\,\Omega$ and $R_{\text{gfe}} \approx 50\,\Omega$. The signal detected at the end of the cable (CH2) is damped and only a small reflected pulse (CH1) can be observed.

If R_{gfe} is removed, the pulse at the end of the cable is almost twice as large as the pulse with an Resistance $R_{\text{gfe}} \approx 50\,\Omega$ at the end of the cable. Since the reflected signal runs back to the beginning, a damped signal is measured there (Fig. 4).

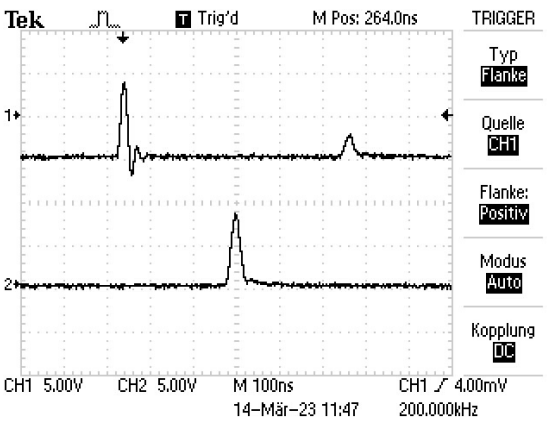

Fig. 4: Detected pulses at both ends of the cable with $R_{\text{gne}} = 50\,\Omega$ and $R_{\text{gfe}} \approx \infty$. The damped signal detected at the end of the cable (CH2) is nearly completly reflected and this reflected pulse (CH1) can be observed.

If the end of the cable is short-circuited, no signal is detected at this end since the pulse is reflected with a phase shift of π. Therefore the incoming and the reflected puls cancels out at the end of the cable. The reflected pulse is now detected again at the beginning, but with a reversed sign of the voltage due to the phase shift (Fig. 5).

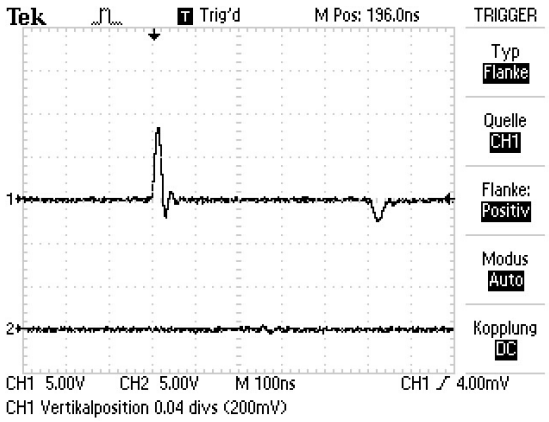

Fig. 5: Detected signals at both ends of the cable with $R_{\text{gne}} = 50\,\Omega$ and a short-circuit plug connected to the generator far end of the cable. No pulse is detected at the end of the cable (CH2), sinse the incomming and the reflected pulse cancel each other out. The inverted and damped reflected pulse is detected at the generator near end (CH1).

R_{gne} is now removed, therefore the beginning of the cable is open ended and the incoming signal is nearly completly reflected at the beginning of the cabel. If R_{gfe} is now switched through again, the same results are observed as in the measurements with $R_{\text{gne}} = 50\,\Omega$, but with a larger amplitude of the detected pulses. An example of this measurement is shown in Fig. 6 for $R_{\text{gfe}} \approx \infty$.
If a short-circuited plug is connected to the beginning of the cable no signal is detected at the generator near and far end, since the pulse is already annihilated at the beginning of the cable.

5.2 Standing waves

For this part a function generator was used, that generates two sine form waves with the same velocity but opposing propagation directions and separatly changeble amplitudes. We connected those signals directly to the input channels of the oscilloscope and plotted both signals together with their superposition.
In Fig. 7 are two incoming waves with the same amplitude shown. This leads to maximum cunstructive interference with the highest maxima and nodes with zero voltage. If one applies incoming waves with different amplitudes U_1 and U_2, the standing wave has a minimum with voltage $|U_1 - U_2|$, as shown in Appendix 9.2. This behavior was confirmed by this ex-

periment.

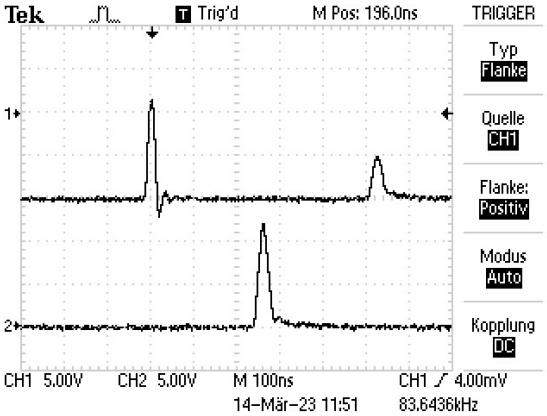

Fig. 6: Detected pulses at both ends of the cable with $R_{\text{gne}} \approx \infty$ and $R_{\text{gfe}} \approx \infty$. The damped signal detected at the end of the cable (CH2) is nearly completly reflected and this reflected pulse (CH1) can be observed. The amplitude of the incomming puls (CH1) is larger than in Fig. 4, since it is reflected at the beginning of the cable.

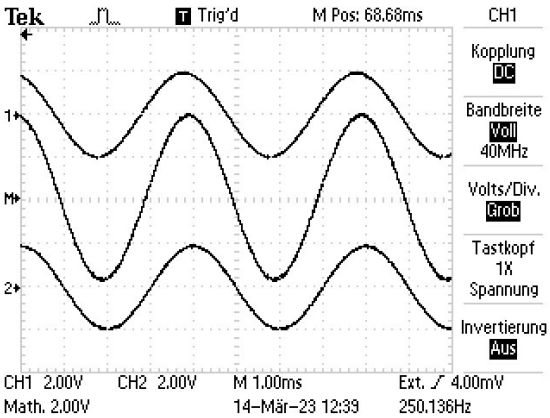

Fig. 7: Standing wave (M) as the superposition of two opposing sine waves (1 and 2) with equal velocity and amplitude.

6 Propagation velocity of waves in a coaxial cable

6.1 Runtime measurement

To determine the propagation velocity in a coaxial cable, one can measure the runtime of a pulsed signal that gets reflected on the open generator far end. We used the experimental setup shown in Fig. (2) with $R_{\text{gne}} = 50\,\Omega$ and $R_{\text{gfe}} = \infty$ to reflect an incoming pulsed signal. We measured the time Δt between the incoming pulse and the outcoming reflected pulse. The length $l = 50\,\text{m}$ of the cable is known. One can calculate the propagation velocity via

$$v = \frac{2l}{\Delta t}. \tag{7}$$

The resulting propagation velocity of the coaxial cable is $v = (1.9077 \pm 0.0087) \cdot 10^8\,\frac{\text{m}}{\text{s}}$ at a frequency of $f = 200\,\text{kHz}$. We did the same measurement with a delay cable with a length of $l = (45.5 \pm 1.0)\,\text{cm}$. We used a low freaquency of $f = 40\,\text{Hz}$ because of the higher damping of this cable. The propagation velocity of the delay cable is $v_{\text{delay}} = (2.132 \pm 0.048) \cdot 10^5\,\frac{\text{m}}{\text{s}}$. The difference in the propagation velocity between both cables can be explained with the different materials. For a standard coaxial cable we aproximated the relative permeability $\mu_{\text{r}} \approx 0$. This approximation could be wrong for the delay cable aud cause the difference.

6.2 Measurement with standing waves

The propagation velocity v of a wave is described by

$$v = f \cdot \lambda \tag{8}$$

where f is the frequency and λ is the wavelength of the wave. To determine the wavelength, one can use standing waves. The nodes of standing waves have a distance of $\frac{\lambda}{2}$. If the length $l = 50\,\text{m}$ of the cable is known, one can calculate λ out of the number of nodes.

Standing waves in a coaxial cable can be produced by a signal that gets reflected at the generator far end. If the generator far end has a resistance $R_{\text{gfe}} = 0$ (short circuit) the standing wave has a node at this end. If the wavelength is exactly $\lambda = 2l$, the first node will be at the generator near end. If the generator far end is open ($R_{\text{gfe}} = \infty$), there will be an antinode at the generator far end. If the wavelength is $\lambda = 4l$ and for $\lambda = \frac{4}{3}l$ there will be the first or second node at the generator near end.

f [MHz]	v [$10^8\,\frac{\mathrm{m}}{\mathrm{s}}$]	ϵ_r	D [dB]
0.97080 ± 0.00086	1.9412 ± 0.0017	2.3841 ± 0.0042	0.615 ± 0.010
1.9439 ± 0.0021	1.9439 ± 0.0021	2.3784 ± 0.0036	0.826 ± 0.033
2.9271 ± 0.0018	1.9514 ± 0.0012	2.3602 ± 0.0019	0.951 ± 0.032

Tab. 2: Propagation velocity v of electromagnetic wave in a coaxial cable depending on its frequency f together with the relative permittivity ϵ_r of the used insulator in the cable and its damping coefficient D.

In Tab. 2 the frequencies f corresponding to the mentioned wavelengths are listed together with the propagation velocity v calculated with eq. (8), the relative permittivity $\epsilon_\mathrm{r} = \frac{c^2}{v^2}$ and the damping coefficient D are listed. The damping coefficient is calculated with

$$D = 20 \cdot \log_{10}\left(\frac{U(0)}{U(0) - U(x)}\right)\,\mathrm{dB} \qquad (9)$$

where $U(0)$ is the amplitude of the incoming wave and $U(x)$ the amplitude of the reflected outcoming wave. One can see that the propagation velocity increases with increasing frequency and the relative permittivity decreases. Both quantities do not change a lot in the observed intervall of frequency. The mean value of the propagation velocity is $v = (1.94742 \pm 0.00089) \cdot 10^8\,\frac{\mathrm{m}}{\mathrm{s}}$. This value does not coincides with the propagation velocity measured via runtime measurement. The mean value of the relative permittivity is $\epsilon_\mathrm{r} = 2.3687 \pm 0,0014$. The used insulator material could be polyethylene, which has a relative permittivity between 2.2 and 2.4 [3].

The damping coefficient includes the fact, that the ohmic resistance of the cable is nonzero. This leads to an exponential decrease of the amplitude of the signal for an increasing distance. This coefficient depends on the frequency of the signal. Lower frequencies are less damped than higher frequencies as we can see in Tab. 2.

7 Measurement of the speed of light in air

For this measurement the experimental setup shown in Fig. 8 was used. A high power LED emits short, red light pulses of a width of about 20 ns. The light pulse is divided by a triple mirror, after which the two pulses then pass through two different paths. One of the two pulses is the reference pulse, which runs a short distance. The other puls runs through a longer variable distance. After running through the two sections, the two pulses are detected and recorded by the oscillograph *Tektronix* TDS 2022C.

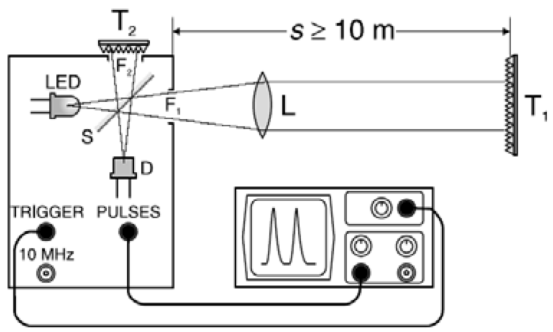

Fig. 8: Experimental setup to measure the speed of light in air. Light pulses are emitted by the LED, which pass throw a triple mirror and so partly run to the reflector T_1 and to the reflector T_2. After the two pulses are reflected, they hit a detector and are recorded by the oscilloscope.

The speed of light is calculated using the distance traveled by the second pulse and the delay of the second pulse, which is determined on the oscillograph. The cursor function of the oscillograph was used to determine the time difference between the reference pulse and the second pulse. The distance traveled as a function of the time required is shown in Fig. 9.
The speed of light was calculated from the slope of the linear fit $c_\mathrm{air} = (2.9421 \pm 0.1205) \cdot 10^8\,\frac{\mathrm{m}}{\mathrm{s}}$. The measured value deviates from the literature value $c_\mathrm{air,lit.} = 299705518\,\frac{\mathrm{m}}{\mathrm{s}}$ by 1.8 % [4]. The good correspondency of the calculated and the literature value can also be seen from the two straight lines in Fig. 9.

8 Conclusion

In this experiment we examined properties of electromagnetic waves in coaxial cables. The characteristic impendance of the used coaxial cable was measured using variable resistors at the generator far end of the cable. Its value is $Z = (49.34 \pm 0.56)\,\Omega$. We also observed the influence of different resistors at both ends of the cable on the incoming and the reflected pulse. If $R_\mathrm{gfe} \approx 50\,\Omega$ one can see a small reflected pulse be-

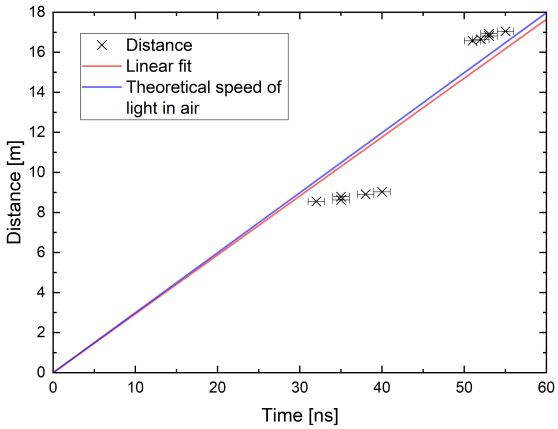

Fig. 9: Determining of the speed of light in air via a runtime measurement. The traveled distance of a light pulse is plotted as a function of the required time and a linear fit is applied to the data. The theoretical course of the speed of light in air is plotted.

cause R_{gfe} has a small deviation from Z. If the cable has an open generator far end, one observes a larger signal at this end. At the generator near end a reflected signal occurs. At a short-circuited generator far end one does not detect a signal. The reflected pulse at the generator near end has a phase shift of π. If the generator near end is open, all detected signal are larger than with $R_{\text{gne}} = 50\,\Omega$. With a short circuit plug connected to the generator near end no signal is detected at both ends. Due to the lower amplitude at the generator far end in comparison to the generator near end, one can observe the damping of the cable. In a simulation experiment one can see the influence of different amplitudes on the superposition of opposing waves.

Furthermore the propagation velocity of waves in a coaxial cable was measured in two different ways. First, we measured the runtime of a reflected pulse in a coaxial cable. The propagation velocity is $v = (1.9077 \pm 0.0087) \cdot 10^8\,\frac{\text{m}}{\text{s}}$. With the same procedure we determined the propagation velocity in a delay cable $v_{\text{delay}} = (2.132 \pm 0.048) \cdot 10^5\,\frac{\text{m}}{\text{s}}$. The second experimental design was to produce standing waves inside the cable. We measured the propagation velocity, the relative permittivity and the damping for three different frequencies. The mean value of the propagation velocity is $v = (1.94742 \pm 0.00089) \cdot 10^8\,\frac{\text{m}}{\text{s}}$ and does not coincides with the first measurement. The mean value of the relative permittivity of the cable is $\epsilon_{\text{r}} = 2,3687 \pm 0,0014$. These values do not depent on the frequency. The damping coefficient increases with higher frequencies from $(0.615 \pm 0.010)\,\text{dB}$ to $(0.951 \pm 0.032)\,\text{dB}$.

We measured the speed of light in air with a runtime measurement of a light pulse in air with different distances. The resulting speed of light in air is $c_{\text{air}} = (2.9421 \pm 0.1205) \cdot 10^8\,\frac{\text{m}}{\text{s}}$.

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[3] `https://www.professionalplastics.com/professionalplastics/ElectricalPropertiesofPlastics.pdf`, last visited 16th March, 2023

[4] https://mathe2go.net/lichtgeschwindigkeit-in-luft-vakuum-wasser-glas-messen-berechnen-tz3/, last visited 20th March, 2023

9 Appendix

In the following paragraphs we will answer the excersises from the instruction [1].

9.1 Numerical example of wave propagation on a coaxial cable

In this section we will compute some physical quantities of a coaxial cable of 50m length that has a capacitance of $C^* = 100\frac{\mathrm{pF}}{\mathrm{m}}$ and an inductance of $L^* = 15 \cdot 10^{-8}\frac{\mathrm{H}}{\mathrm{m}}$. The relation between the outer and the inner diameter is $d_\mathrm{a} = 3d_\mathrm{i}$.
The time the signal takes to travel through the cable is

$$\Delta t = \frac{\Delta x}{v} = \Delta x\sqrt{L^* C^*} = 0,35\mathrm{ms}.$$

The relative permittivity of the insulator in the cable is

$$\epsilon_\mathrm{r} = L * C * c^2 = 2,25.$$

If we assume $\mu_\mathrm{r} = 1$, we can calculate the impendanceof this cable with the previous results:

$$Z = \sqrt{\frac{\mu_\mathrm{r}\mu_0}{\epsilon_\mathrm{r}\epsilon_0}}\frac{1}{2\pi}\ln\frac{d_\mathrm{a}}{d_\mathrm{i}} = 43,94\Omega$$

If we solve this equation for $\frac{d_\mathrm{a}}{d_\mathrm{i}}$, we can compute the relation of the diameters for a cable with impendace $Z = 50\Omega$:

$$\frac{d_\mathrm{a}}{d_\mathrm{i}}(Z = 50\Omega) = 3,49$$

9.2 Mathematical describtion of waves

The simplest form af a wave is the harmonic monochromatic wave. It can be described by

$$y(x,t) = a \cdot \sin 2\pi\nu\big(t - \frac{x}{c}\big). \tag{10}$$

x is the position, t the time, a the amplitude, ν the frequency and c the propagation velocity of the wave. The wavelegth λ is the smallest length fulfilling the the condition $y(x,t) = y(x+\lambda,t), \forall x,t$. by applying this condition to the wave equation (10), you get the important relation

$$c = \lambda\nu. \tag{11}$$

Adding up two waves of the form (10) with the same amplitudes a but different frequencies ν_1 and ν_2 and opposing propagation directions, we get a resulting wave

$$\begin{aligned} y &= y_1 + y_2 \\ &= 2a\sin\Big(2\pi\nu^*\big(t - \frac{x}{c}\big)\Big)\cos\Big(2\pi\Delta\nu\big(t - \frac{x}{c}\big)\Big) \end{aligned} \tag{12}$$

with $\nu^* = \frac{\nu_2+\nu_1}{2}$ and $\Delta\nu = \frac{\nu_2-\nu_1}{2}$. This represents a wave with an amplitude $2a$, a frequency ν^* ans an envelope with a frequency $\Delta\nu$. At the nodes of the envelope the amplitude vanishes completely.
When looking at the superposition of two waves in form of (10) with the same frequency ν but different amplitudes a_1 and a_2 and opposing propagation directions, we get

$$\begin{aligned} y &= y_1 + y_2 \\ &= (a_1 + a_2)\cos\left(2\pi\nu\frac{x}{c}\right)\sin(2\pi\nu t) \\ &+ (a_2 - a_1)\sin\left(2\pi\nu\frac{x}{c}\right)\cos(2\pi\nu t). \end{aligned} \tag{13}$$

This is the mathematical representation of a standing wave with the amplitude oszillating between $a_1 + a_2$ and $|a_2 - a_1|$ and a separation in phase of $\frac{\pi}{2}$.
In the previous paragraphs we only looked at waves with linear dispersion. In general the propagation velocity depends on the wavelength $c(\lambda)$. Looking back to eq. (12) and implementing those different velocities $c_1 = c(\lambda_1$ and $c_2 = c(\lambda_2) = c_1 - \Delta c$, $\lambda_2 = \lambda_1 - \Delta\lambda$ and referring to very small deviations in velocity and wavelegth ($\Delta c \gg c_1$ and $\Delta\lambda \gg \lambda_1$), we get

$$\begin{aligned} y =& 2a\cos\left(\pi\frac{(\lambda_1\Delta c - \Delta\lambda c_1)\cdot t + \Delta\lambda\cdot x}{\lambda_1^2}\right) \\ &\cdot\sin\left(2\pi\frac{c_1\cdot t - x}{\lambda_1}\right). \end{aligned} \tag{14}$$

For comparison with eq. (12), we look at the limit $\Delta c \to 0$, aproxximate $\lambda_1\lambda_2 \approx \lambda_1^2$ and implementing eq. (11) in the form $\nu_\mathrm{i} = \frac{c_1}{\lambda_\mathrm{i}}$. This exactly reproduces eq. (12).